ASSIGNMENT PROBLEMS

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Assignment Method

- Involves tasks or jobs to resources.
- It is a way of allocating organizational resources in which each resource is assigned to a particular task. The resource could be monetary or technological.
- Assignment method is used to determine what resources are assigned to which department, machine, or center of operation in the production process.
- The objective is to minimize total costs or time required to perform the tasks at hand. It should enhance production efficiency

Assignment Method examples

- The assignment method has various applications in maximizing resources, including:
  - Allocating the proper number of employees to a machine or task.
  - Allocating a machine or a manufacturing plant, number of jobs that a given machine or factory can produce.
  - Assigning new computers, laptops, tablets and other expensive high-tech devices to the areas that need them the most. Lower priority departments may get the older models.
  - Assigning a number of salespersons to a given territory or territories.

- ***Example:*** 3 different 3D printers (Printer 1, 2 and 3) are available at a factory. 3 products (A, B and C) are to be fabricated. Determine the optimal assignment that minimizes the cost.

	Different 3D printers		
Product	Printer 1	Printer 2	Printer 3
A	Rs 120	Rs 140	Rs 60
B	Rs 80	Rs 100	Rs 110
C	Rs 90	Rs 120	Rs 70

- The assignment method involves adding and subtracting appropriate numbers in the table to find the lowest ***assignment (optimal) cost***. It is the value of what one lose when choosing between two or more options. The steps are as follows:

1. Subtract the smallest number in each row from every number in that row. Then, from the resulting matrix, subtract the smallest number in each column from every number in that column. This step has the effect of reducing the numbers in the table until a series of zeros.

2. Draw the minimum number of vertical and horizontal straight lines to cover all zeros in the table. If the number of line equals either the number of rows or the number of columns in the table, optimal assignment can be made. If the number of lines is less than the number of rows or columns, proceed to Step 3.
3. Subtract the smallest number not covered by a line from every other uncovered number. Add the same number to any number(s) lying at the intersection of any two lines. Return to Step 2 and continue until an optimal assignment is possible.
4. Optimal assignments will always be at zero locations in the table. Select a row or column that contains only one zero square. Assignment can be made to that box and then draw lines through its row and column. From the uncovered rows and columns, choose another row or column in which there is only one zero. Now, assignment is made and the procedure is continued until assigned each person or machine is assigned a task.

	Different 3D printers		
Product	Printer 1	Printer 2	Printer 3
A	120	140	60
B	80	100	110
C	90	120	70

Subtract the smallest number from each row.

	Different 3D printers		
Product	Printer 1	Printer 2	Printer 3
A	60	80	0
B	0	20	30
C	20	50	0

Subtract the smallest number from each column.

Draw minimum number of vertical and horizontal lines needed to cover all zeros as shown below.

	Different 3D printers		
Product	Printer 1	Printer 2	Printer 3
A	60	60	0
B	0	0	30
C	20	30	0

Subtract the smallest uncovered number (20 in the table on the right side) from every uncovered number. Also, add it to numbers at the intersection of two lines

	Different 3D printers		
Product	Printer 1	Printer 2	Printer 3
A	40	40	0
B	0	0	50
C	0	10	0

Select the row or column that contains only zero. Mark that zero by square bracket and draw a line in the row and column covering that zero.

	Different 3D printers		
Product	Printer 1	Printer 2	Printer 3
A	40	40	[0]
B	0	0	50
C	0	10	0

From the uncovered rows and columns, choose another row or column to where there is only one zero.

	Different 3D printers		
Product	Printer 1	Printer 2	Printer 3
A	40	40	[0]
B	0	[0]	50
C	[0]	10	0

- Optimal assignment can now be made.
- Product A should be fabricated in Printer 3,

  Product B in Printer 2 and Product C in Printer 1.
- Minimum cost is Rs. (60+100+90) = Rs. 250

	Different 3D printers		
Product	Printer 1	Printer 2	Printer 3
A	120	140	60
B	80	100	110
C	90	120	70

Three Ph.D. students X, Y and Z have to be assigned to three supervisors A, B and C. Estimated number of papers each pair can produce in 5 years are given in the form of matrix. How should I assign students to supervisors, such that department publishes the maximum number of papers?

	A	B	C
X	15	10	9
Y	9	15	10
Z	10	12	8

This is a classic **assignment problem**, which can be solved using the **Hungarian algorithm** (or other optimization techniques like integer programming). The goal is to maximize the number of papers published by assigning students to supervisors in the best possible way.

Apply the Hungarian Algorithm (Maximization Case)

- Convert the problem into a **minimization problem** by subtracting each element from the **maximum element** in the matrix.
- Solve the transformed problem using the Hungarian algorithm, which finds the optimal assignment.
- Otherwise:
- The Hungarian Algorithm solves minimization problems, so we convert the given matrix into a cost matrix by negating all values.

Transformed matrix

-15	-10	-9
-9	-15	-10
-10	-12	-8

- For each row, subtract the smallest value in that row from all elements in the row:

0	5	6
6	0	5
2	4	0

- For each column, subtract the smallest value in that column from all elements in the column:

0	5	6
6	0	5
0	4	0

- Using the Hungarian Algorithm's zero-covering method, we determine the optimal assignment:
  - - Student X → Supervisor A
  - - Student Y → Supervisor B
  - - Student Z → Supervisor C
- Total maximum papers published: 38

A manufacturing company has four jobs (J1, J2, J3, J4) that need to be assigned to four machines (M1, M2, M3, M4). Each job must be assigned to exactly one machine, and each machine can handle only one job. The time taken (in hours) for each job on each machine is given in the following cost matrix:

	M1	M2	M3	M4
J1	12	7	9	11
J2	8	5		7
J3	6	8	10	9
J4	7	6	9	8

Note that machine M3 can do job J2.

Use the Hungarian Method to determine the optimal assignment of jobs to machines such that the total time is minimized.

Assuming a large value in place of missing time so that it will become unsuitable for assignment.

	M1	M2	M3	M4
J1	12	7	9	11
J2	8	5	100	7
J3	6	8	10	9
J4	7	6	9	8

	M1	M2	M3	M4
J1	5	0	2	4
J2	3	0	95	0
J3	0	2	4	3
J4	1	0	3	2

	M1	M2	M3	M4
J1	5	0	0	2
J2	3	0	93	0
J3	0	2	2	1
J4	1	0	1	0

	M1	M2	M3	M4
J1	5	0	X0	2
J2	3	X0	93	0
J3	X0	2	2	1
J4	1	0	1	X0

Minimum total time = 9+5+6+8 = 28 hours

A company needs to assign 3 workers (A, B, and C) to 3 different jobs (X, Y, and Z). The cost (in rupees) of assigning each worker to each job is given in the matrix on the next page:

	X	Y	Z
A	1200	700	900
B	800	600	500
C	700	800	600

To determine the optimal assignment using the **Hungarian method**, follow these steps:

Step 1: Construct the Cost Matrix

Given the cost matrix:

	X	Y	Z
A	1200	700	900
B	800	600	500
C	700	800	600

Step 2: Row Reduction

Subtract the minimum value in each row from all elements in that row.

- Row A: Minimum = **700**

$$(1200 - 700, 700 - 700, 900 - 700) = (500, 0, 200)$$

- Row B: Minimum = **500**

$$(800 - 500, 600 - 500, 500 - 500) = (300, 100, 0)$$

- Row C: Minimum = **600**

$$(700 - 600, 800 - 600, 600 - 600) = (100, 200, 0)$$

New reduced matrix:

	X	Y	Z
A	500	0	200
B	300	100	0
C	100	200	0

Step 3: Column Reduction

Subtract the minimum value in each column from all elements in that column.

- Column X: Minimum = **100**

$$(500 - 100, 300 - 100, 100 - 100) = (400, 200, 0)$$

- Column Y: Minimum = **0** (no change)
- Column Z: Minimum = **0** (no change)

New reduced matrix:

	X	Y	Z
A	400	0	200
B	200	100	0
C	0	200	0

Step 4: Covering Zeros with Minimum Number of Lines

- Mark one zero in each row and column while ensuring minimum number of lines.
- Assign jobs optimally:
  - Assign C → X (0)
  - Assign A → Y (0)
  - Assign B → Z (0)

Step 5: Optimal Assignment & Minimum Cost

The assignments are:

- **C → X** (700)
- **A → Y** (700)
- **B → Z** (500)

Total minimum assignment cost:

$$700 + 700 + 500 = 1900$$

Final Answer:

- **Optimal Assignment:**
  - **A → Y**
  - **B → Z**
  - **C → X**
- **Minimum Assignment Cost = ₹1900**